

Search corridor

Revise 360 lesson plan — 60 minutes

Unit	OCR J277 2.1 Algorithms — Paper 2, Computational thinking, algorithms and programming
Lesson	Search corridor (lesson 6 of the unit)
Lesson ID	a1-l06
Experience ID	a1-l06
Plan ID	PLAN-AL-L06
Content version	68abe2d3498e
Website	https://revise360.co.uk/experience.html?id=a1-l06

Big question, outcomes and success criteria

Big question — How does a linear search work, and when is it the right choice?

Prerequisites

The lesson before this one is a1-l05. Pupils should already have finished it: the starter below retrieves from it.

Earlier in the unit: a1-l01, a1-l02, a1-l03, a1-l04.

Assessable outcomes

1. **a1-l06-o1** — understand the linear search algorithm
2. **a1-l06-o2** — understand that it is inefficient but simple and needs no ordering

Success criteria the pupil can check against

Taken from the briefs the independent activities themselves show, so these are the words the pupil sees. Their work has met the outcome when it does these things.

- The list is already written for you - do not change it. a1-l06-s5-a3

Vocabulary, resources and preparation

Key vocabulary — Linear search, Worst case

Resources

SLIDES	presentations/AL_L06_SearchCorridor_Lesson.pptx
WORKSHEET	worksheets/AL_L06_SearchCorridor_Worksheet.docx
ONLINE EXPERIENCE	experience.html?id=a1-l06

Before the lesson

1. Sign in on the **teacher dashboard** (teacher.html) and open **Class logins**. Enter the class or group and the usernames, one per line, then print the login cards. A pupil cannot sign in without one, so do this before the lesson rather than in it.
2. Print the worksheet, one per pupil.
3. Open the slides and the experience once yourself, on the machine you will be teaching from, so the 360 image is already cached.
4. Check headsets or devices are charged and on the school network.

How pupils get in

1. Go to `topics.html` on `revise360.co.uk`.
2. Sign in with the **username and PIN from the login card**. The class is already set against the card.
3. Choose the topic, then this lesson's card.

Worth knowing before you promise anything. In this build `APP_CONFIG.backendUrl` is empty, so logins, login cards and progress are stored on the device the pupil used. The teacher dashboard says so too. A pupil who moves to a different machine starts again, so keep a class on the same devices within a unit, and treat the paper worksheet as the record that travels.

Teaching sequence

Sixty minutes including transitions. The online block is the one to move if your lesson is shorter: see the adaptation at the end.

MIN	PHASE	TEACHER	PUPILS	EVIDENCE TO CHECK
0-5	Retrieve	Starter on the board or the worksheet, from the last lesson and this lesson's prerequisites.	Answer the starter on the worksheet.	The starter box on <code>AL_L06_SearchCorridor_Worksheet.docx</code> is filled in.
5-8	Review the starter	Take answers; correct the one most got wrong.	Correct their own starter.	A correction written, not just a tick.
8-18	Explain and model	The new idea in two or three steps, with the example worked in front of the class.	Listen and answer questions about the example.	They can say why, not only what.
18-21	Set up and settle	Sign-in, launch, and what to write before anyone taps an answer.	Sign in, open the lesson, find How it works.	Everyone is in and on station 1.
21-43	Online: work the stations in order	Pupils read each wall, write the key fact and the challenge, then answer at the badge.	Read each wall, write the key fact and the challenge, then answer at the badge.	The worksheet has writing on it for each of the 6 stations before the badge is tapped.
43-48	Check understanding	The hinge question, hands down.	Answer the hinge question, hands down.	The split across the options.
48-56	Apply	Exam practice on the worksheet, in silence.	Exam practice, in silence.	worksheet, Exam practice a) to c)
56-60	Exit and set up the next lesson	The exit question and the retrieval link to next lesson.	Write the exit answer.	One sentence that answers the big question.

The explanation and the worked example

This is what the walls of the experience say, station by station. Teach the first two or three from the front before anyone puts a headset on; the rest the pupils read for themselves.

1. How it works

- Start at the first item and compare it to what you are looking for.
- If it doesn't match, move to the next and compare again.
- Stop when it matches, or when you reach the end without finding it.

Challenge on this wall: Describe a linear search in three steps of your own.

2. Its one great advantage

- The list does not need to be in order.
- It works on any data that can be compared, including text.
- It is short and easy to program, so mistakes are less likely.

Challenge on this wall: Give two situations where a linear search is the right choice.

3. Counting the checks

- Best case: the item is first, so one check.
- Worst case: the item is last or missing, so one check per item.
- On average, about half the list is checked.

Challenge on this wall: For a list of 100 items, state the best, worst and average number of checks.

4. Why it gets slow

- Double the list and you double the work.
- For a million items, the worst case is a million comparisons.
- This is why large systems sort their data and use a faster search instead.

Challenge on this wall: Explain why a linear search becomes unusable on very large data.

5. Writing it down

- Set a flag to false, then loop through every index.
- If the item at that index matches, set the flag to true and stop.
- After the loop, the flag says whether it was found.

Challenge on this wall: Write pseudocode for a linear search that outputs true or false.

6. Answering search questions

- Name the algorithm and say what it does step by step.
- Say whether the list must be in order: for a linear search, it need not be.
- If asked to count checks, count comparisons, not loops.

Challenge on this wall: Plan a four-mark answer describing a linear search.

Retrieval and the check for understanding

Starter — You have lost a photo in an album of 200 pictures. Describe exactly how you would look for it, and how many you would expect to check.

Answer space: 3 lines on the worksheet.

The answer is in the teacher answers for this unit, not on this plan: a plan left on a desk should not give it away.

Hinge question

Use this one, hands down, before anyone moves on to independent work. It is a real activity from the experience (a1 - 106 - s4 - c1), so the wording a pupil sees matches the wording you say.

A list grows from 1,000 to 2,000 items. What happens to the worst case?

Correct: It doubles, to 2,000 checks

What a pupil might choose instead

- It stays the same
- It goes up by one
- It quadruples

Count the hands on each. If more than a few of the class are on one wrong option, that option is the misconception to teach against before anyone goes on: go back to the station this question sits on and work through its example again, aloud, before the independent work.

The course's own response to a wrong answer here: The work grows in step with the list, which is what makes it unsuitable for huge data.

One response, not four. one response covers every wrong option; it does not say what to revisit for each one, so decide your own reteach for each option before the lesson - the list above is the start of it.

Guided to independent practice

STAGE	ACTIVITIES	WHAT IT IS
Guided	11	Support is in the activity: an example to run, an output to predict, one thing to change, a gap to fill.
Independent	4	The pupil writes or fixes it with the brief and nothing else.
Challenge	0	The hardest activity on each station. It is not optional and it is not extra: it is the last step of the station.

Scaffolds already in the product, and how they fade

- **Hint** is a ladder, not a door: the first rung names the idea, the next shows the shape of the code, the next how it starts, the last animates the technique on different data. A pupil asks for each rung. None of them is the answer.
- **Run** is free and unmarked; **Check my answer** marks. A pupil may run as often as they like, which is what makes trial and error safe.
- **I need help** says to ask you, and says plainly that the question still has to be completed. It does not notify anyone and it does not unlock anything.
- The worked example appears on every early activity and on none of the late ones. That fade is deliberate; it is not a missing example.

Accessibility

- **Read aloud** on every task window reads the task and its brief, with the prescribed values read as values rather than as punctuation.
- Text is 16px or larger throughout, and a question window scrolls rather than shrinking its type.
- Prescribed data is boxed and in a monospace face as well as coloured, so it still reads as code in black and white, in a high-contrast mode, and to a pupil who does not separate green from amber.
- **Lower literacy and EAL:** pre-teach the vocabulary above; let the pupil say the key fact before writing it; accept the key fact in two or three words.
- **SEND:** the sequence is compulsory but the pace is not. A pupil may finish a station next lesson; the progress saves after every answer and reopens at their earliest unfinished question.
- A headset is never required. Every experience works on a laptop or a tablet with a mouse or a finger.

Delivering the online experience

Before anyone enters

- Worksheet out, pen in hand. The rule is write first, then answer at the badge: the first answer is the one that counts.
- Say where the stations are. They are numbered around the room: 1 and 2 to the right, 3 and 4 behind, 5 and 6 to the left, and the final challenge on the floor.
- Say that a badge marked **Not yet** is not a fault. It opens when the station they are on is finished.

The stations, and what to record at each

#	STATION	ACTIVITIES	RECORD ON PAPER
1	How it works	2	A linear search works by...
2	Its one great advantage	2	A linear search is useful because...
3	Counting the checks	2	A linear search needs...
4	Why it gets slow	2	A linear search gets slower because...
5	Writing it down	3	Pseudocode for a linear search...

#	STATION	ACTIVITIES	RECORD ON PAPER
6	Answering search questions	2	To answer a search question, I should...
final	Final challenge: search it	2	the answer to the station's challenge

Where to pause

Stop the class after station 2 and take answers to its challenge out loud. That is the point where a pupil who has misread the technique can still be put right cheaply.

Afterwards

Headsets off and down. Ask three pupils for the key fact from a station of your choosing, in their own words and not read off the sheet. Then the exit question.

Assessment, evidence and unfinished work

ID	QUESTION	MARKS	WHERE
a1-106-x1	Describe how a linear search works.	3	worksheet, Exam practice a); deck, Exam practice slide
a1-106-x2	State one advantage of a linear search over a binary search.	1	worksheet, Exam practice b); deck, Exam practice slide
a1-106-x3	A list contains 2,000 unsorted items. Explain why a linear search may take a long time, and state the maximum number of comparisons needed.	4	worksheet, Exam practice c); deck, Exam practice slide

Expected answers

Model answers and marking guidance are teacher-only and are not printed on this plan. They are in the unit's teacher guide pack, and the per-activity marking for this lesson is in `answers/records/a1-106.json`, which is not published with the website.

Common errors

The wrong options the activities offer are the errors this lesson expects. The commonest to watch for:

- **Where does a linear search start?** — At the middle item; At the last item
- **What is the main advantage of a linear search?** — It is the fastest search; It uses no memory
- **A list has 500 items. What is the worst case for a linear search?** — 1 check; 9 checks
- **A list grows from 1,000 to 2,000 items. What happens to the worst case** — It stays the same; It goes up by one
- **Why do large systems sort their data?** — Sorted data uses less space; Unsorted data cannot be stored

Unfinished work

Record where each pupil stopped, not a tick. They resume at their earliest unfinished question, and a station left half-done is the thing to open next lesson. Do not mark the lesson complete for a pupil who did not finish it: the score screen already shows what was and was not done, and that is the honest record.

Exit, next lesson and homework

Exit check

One of these, written, not a show of hands:

- How a linear search works
- Its advantages and disadvantages
- Counting the comparisons

The confidence columns on the worksheet are a supplement to this, not a substitute: a pupil who rates themselves confident and cannot answer the exit question has told you nothing.

Retrieval link to next lesson

Next is a1-107, Halving hall. Use this lesson's big question as next lesson's starter, so the first thing they do is retrieve it.

Homework

Finish any station left open, on the website, from the question they stopped at. If the class has no devices at home, set the exam practice on the worksheet instead and mark it next lesson.

Shorter lessons, longer lessons and no internet

A 40 to 45 minute lesson

Keep the starter, the explanation and the model at full length: cutting those is what makes the online work fail. Take the time out of the online block and the apply phase. Pupils stop where they are; their progress saves after every answer and reopens at their earliest unfinished question, so the next lesson starts there.

If the internet or the devices are unavailable

Teach from the slides and the worksheet. The slides carry the same teaching as the station walls, and the worksheet carries the key facts, the challenges and the exam practice. Set the online stations when the devices are back.

There is no offline version of the experience, and this plan does not contain one. Do not hand out a printed walkthrough of the stations as a substitute: the experiences are used on the website.

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